

1. What are Life Processes?

Definition: Maintenance activities essential for living organisms to sustain life on Earth.

Single-Celled vs Multicellular Organisms:

- **Single-Celled (Amoeba):** Entire body surface is in direct contact with environment; simple diffusion is sufficient for gas exchange & nutrient intake.
- **Multicellular (Humans):** Cells are not in direct contact with environment; simple diffusion is too slow to meet tissue oxygen demand. Specialized organ systems are required.

2. Autotrophic Nutrition (Photosynthesis)

Organisms synthesize organic food (glucose) from simple inorganic raw materials (CO_2 & H_2O) using sunlight and chlorophyll.

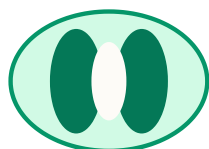


3 Essential Steps of Photosynthesis (CBSE Direct 3-Mark Question):

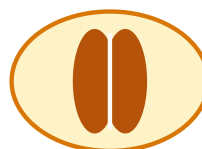
1. **Absorption** of light energy by chlorophyll.
2. **Conversion** of light energy to chemical energy & **photolysis** (splitting) of water into H_2 and O_2 .
3. **Reduction** of Carbon Dioxide into Carbohydrates (Glucose).

Stomata & Guard Cell Mechanism

Guard Cells regulate opening/closing of stomatal pores. When guard cells absorb water, they *swell* and curve outwards to open the pore. When water is lost, they *shrink* and close the pore.



Open Stoma (Swollen)



Closed Stoma (Shrunk)

Figure 1.1: Guard Cell Regulation during Gas Exchange

3. Human Digestive System & Enzymes

Food is broken down mechanically and chemically as it moves through the alimentary canal.

Organ / Gland	Secretion / Enzyme	Primary Action / Function
Mouth	Salivary Amylase	Breaks down complex starch into simple sugar (maltose).
Stomach	Gastric Juice (HCl, Pepsin, Mucus)	HCl provides acidic pH and kills microbes; Pepsin digests proteins; Mucus protects stomach wall.
Liver	Bile Juice	Emulsifies large fat globules into small droplets; makes medium alkaline.
Pancreas	Trypsin & Lipase	Trypsin digests proteins; Lipase breaks down emulsified fats.
Small Intestine	Intestinal Enzymes	Final conversion: Carbs → Glucose, Proteins → Amino Acids, Fats → Fatty Acids + Glycerol.

Villi: Microscopic finger-like projections in the small intestine lined with capillaries that maximize surface area for rapid nutrient absorption.

4. Respiration & Glucose Breakdown Pathways

Respiration is the cellular process where glucose is oxidized to generate usable energy in the form of **ATP**.

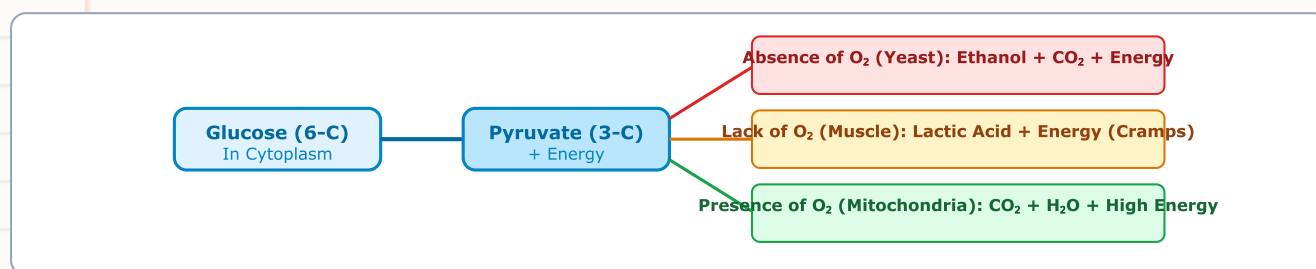


Figure 2.1: Three Pathways of Glucose Breakdown

5. Transportation in Humans & Plants

Human Circulation & Double Loop

Humans have 4-chambered hearts ensuring complete separation of oxygenated and deoxygenated blood to maintain constant warm-blooded body temperature.

Blood Vessel	Wall Thickness	Direction & Pressure	Valves
Arteries	Thick & Elastic	Away from heart (High pressure)	Absent
Veins	Thin	Towards heart (Low pressure)	Present (prevents backflow)
Capillaries	1-cell thick	Site of tissue exchange	Absent

Plant Transportation Comparison

Xylem: Transports water & minerals unidirectionally using *transpiration pull* (day) and *root pressure* (night).

Phloem: Transports photosynthetic sugars (sucrose) bidirectionally (**Translocation**) using cellular ATP energy.

6. Human Excretory System & Nephron

The **Nephron** is the structural and functional filtration unit of the kidney.

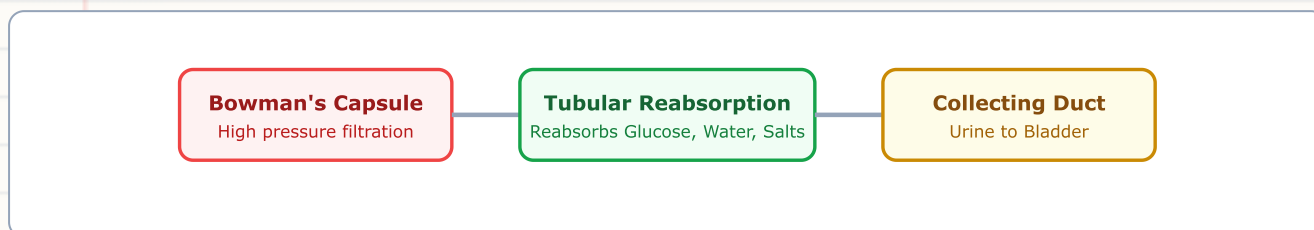


Figure 3.1: Selective Reabsorption Process in Nephron

Hemodialysis (Artificial Kidney): Procedure used to filter nitrogenous metabolic wastes from the blood of patients suffering from kidney failure.